

SENSE AI

Predict the vibe before the credits roll.

AI-Based Movie Review Sentiment Analysis

Using Deep Learning & Natural Language Processing

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Problem Statement



Volume Overload

Millions of movie reviews are posted daily across IMDb, Rotten Tomatoes, and social media — impossible to read manually.



Time & Efficiency

Manual review analysis is slow, inconsistent, and scales poorly for studios and streaming platforms.



The Need

An automated AI system that reads reviews and instantly determines whether audience sentiment is Positive or Negative.

Project Overview & Objectives

Project Overview

This project develops an AI system that automatically analyzes IMDB movie reviews and classifies sentiment as Positive or Negative.

It uses NLP techniques and deep learning to process 50,000 reviews, then deploys the model as a Streamlit web application called SENSE AI.

Project Goals

- ▶ Apply NLP techniques for text preprocessing
- ▶ Train deep learning models (SNN, CNN, LSTM)
- ▶ Compare model architectures for best accuracy
- ▶ Build an automated movie review analyzer
- ▶ Deploy as a user-friendly web application

Dataset & NLP Preprocessing

Dataset

 IMDB Movie Reviews

 50,000 reviews

✓ 25,000 Positive

✗ 25,000 Negative

 80/20 Train-Test Split

NLP Preprocessing Pipeline

1

Lowercasing

Convert all text to lowercase for uniformity

2

HTML Tag Removal

Strip HTML tags from raw review text

3

Punctuation Removal

Remove special characters and numbers

4

Stopword Removal

Filter common words (NLTK stopwords)

5

Tokenization

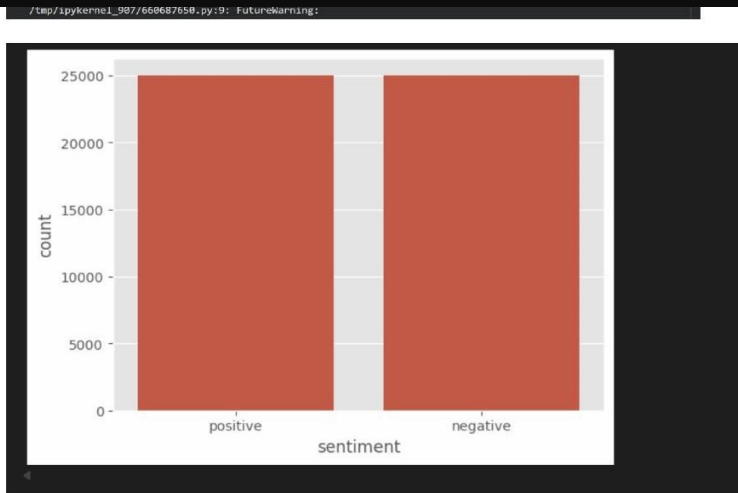
Convert words to numerical sequences (Keras)

6

Sequence Padding

Pad all sequences to fixed length of 100

Dataset — Sentiment Distribution



50,000

Total Reviews

25K

Positive

25K

Negative

Perfectly balanced dataset ensures unbiased training.

Model Architectures

SNN

Simple Neural Network

- ▶ Embedding Layer
- ▶ Flatten Layer
- ▶ Dense (64, ReLU)
- ▶ Dense (1, Sigmoid)

Baseline model — fast training, simpler structure

CNN

Convolutional Neural Network

- ▶ Embedding Layer
- ▶ Conv1D (128 filters)
- ▶ GlobalMaxPool1D
- ▶ Dense (1, Sigmoid)

Captures local patterns and n-gram features

LSTM

Long Short-Term Memory

- ▶ Embedding Layer
- ▶ LSTM (128 units)
- ▶ Dropout (0.3)
- ▶ Dense (1, Sigmoid)

Best performer — understands word order & context

All models use pre-trained GloVe 6B 100d word embeddings for semantic feature representation.

System Workflow



Training Configuration

Epochs: **10**

Batch Size: **128**

Validation Split: **20%**

Optimizer: **Adam**

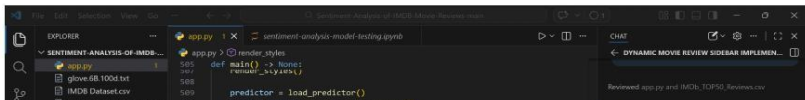
Loss Function: **Binary Crossentropy**

Sequence Length: **100 tokens**

Training Progress — LSTM Model

MODEL ACCURACY INCREASING AFTER EACH ITERATION

```
Epoch 1/10  
250/250 — 43s 165ms/step - accuracy: 0.7440 - loss: 0.5214 - val_accuracy: 0.7989 - val_loss: 0.46  
Epoch 2/10  
250/250 — 40s 161ms/step - accuracy: 0.8136 - loss: 0.4194 - val_accuracy: 0.8378 - val_loss: 0.38  
Epoch 3/10  
250/250 — 40s 159ms/step - accuracy: 0.8385 - loss: 0.3740 - val_accuracy: 0.8404 - val_loss: 0.37  
Epoch 4/10  
250/250 — 40s 159ms/step - accuracy: 0.8525 - loss: 0.3480 - val_accuracy: 0.8575 - val_loss: 0.34  
Epoch 5/10  
250/250 — 41s 162ms/step - accuracy: 0.8659 - loss: 0.3203 - val_accuracy: 0.8599 - val_loss: 0.32  
Epoch 6/10  
250/250 — 40s 162ms/step - accuracy: 0.8746 - loss: 0.3039 - val_accuracy: 0.8671 - val_loss: 0.31  
Epoch 7/10  
250/250 — 41s 163ms/step - accuracy: 0.8798 - loss: 0.2926 - val_accuracy: 0.8651 - val_loss: 0.31  
Epoch 8/10  
250/250 — 45s 179ms/step - accuracy: 0.8865 - loss: 0.2790 - val_accuracy: 0.8579 - val_loss: 0.31  
Epoch 9/10  
250/250 — 45s 181ms/step - accuracy: 0.8977 - loss: 0.2564 - val_accuracy: 0.8652 - val_loss: 0.30  
Epoch 10/10  
250/250 — 39s 156ms/step - accuracy: 0.9046 - loss: 0.2434 - val_accuracy: 0.8712 - val_loss: 0.30
```



Key Metrics

74.4%

Epoch 1 (acc)

86.6%

Epoch 5 (acc)

90.5%

Epoch 10 (acc)

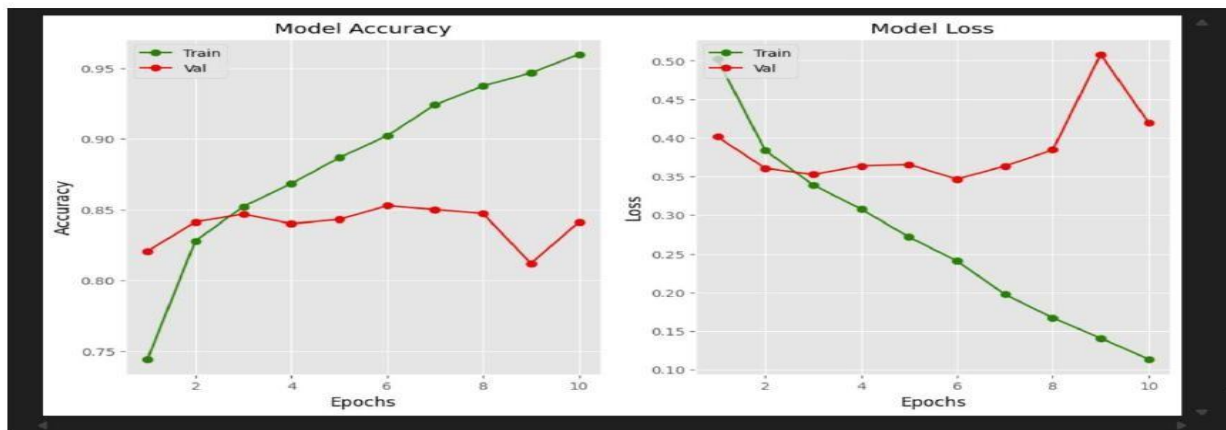
↓ 0.24

Val Loss (final)

87.1%

Val Acc (final)

Model Accuracy & Loss Curves



The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code cell at the top. The code cell contains the following Python code:

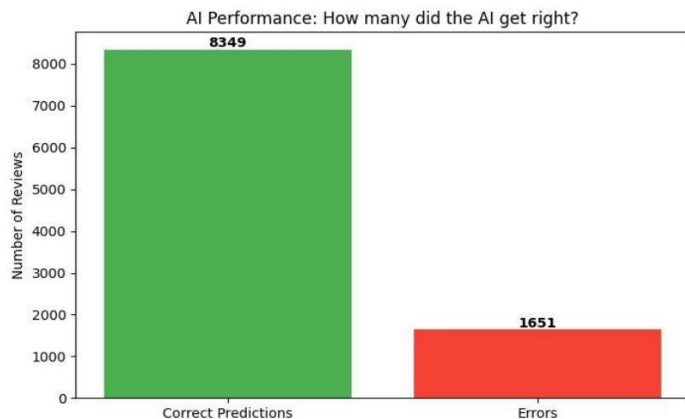
```
pred_df = df.drop(columns=["Unnamed: 0"], errors="ignore")
```

Below the code cell, there is a table with 5 columns: 'Movie', 'Review Text', 'IMDb Rating', 'Predicted Sentiment', and 'Predicted Review Sentiment'. The table contains two rows of data.

	Movie	Review Text	IMDb Rating	Predicted Sentiment	Predicted Review Sentiment
0	The Shawshank Redemption	A profound masterpiece about the resilience of...	10	9.9	Positive
1	The Godfather	The pinnacle of crime drama. Coppola's direct...	10	9.9	Positive

Training accuracy steadily increases to 90%+ while loss decreases — confirming the LSTM model learned well over 10 epochs.

Model Performance Results



data-sentiment_classifierh5 Deleted
From <https://www.kaggle.com>

8,349

Correct Predictions

1,651

Errors

~83.5% Accuracy on 10,000 test reviews

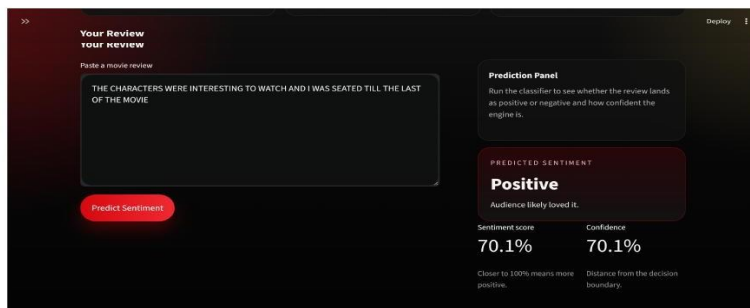
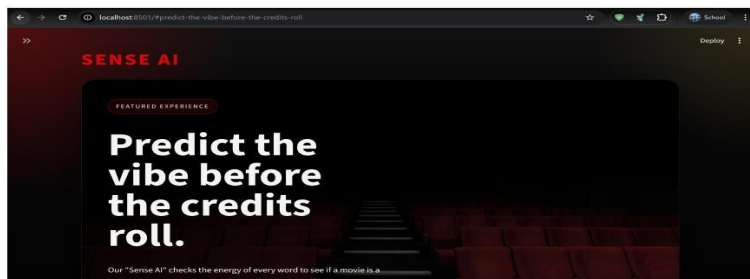
Sample Prediction Results



2	The Dark Knight	Nolan redefined the superhero genre. Ledger's ...	9	9.8	Positive
3	Breaking Bad	The greatest character study ever televised. W...	10	9.9	Positive
4	The Sopranos	Revolutionized television. James Gandolfini's ...	10	9.9	Positive
5	Band of Brothers	A visceral, deeply moving tribute to the men w...	10	9.9	Positive
6	Chernobyl	A chilling, claustrophobic look at human error...	9	9.9	Positive
7	Avatar: The Last Airbender	Perfect storytelling for all ages. It balances...	10	9.9	Positive
8	The Wire	A gritty, uncompromising look at the American ...	9	9.9	Positive
9	Game of Thrones	Despite a divisive ending, the first six seaso...	8	9.9	Positive
10	Interstellar	Visually stunning and emotionally heavy. Nolan ...	9	9.9	Positive

LSTM model correctly predicted sentiment for top-rated movies including Shawshank Redemption, The Dark Knight, and Breaking Bad.

SENSE AI — Streamlit Web Application



App Highlights

- ✓ Netflix-inspired dark UI design
- ✓ Real-time sentiment prediction
- ✓ Confidence score display
- ✓ IMDb sample review catalog
- ✓ Sidebar movie selector
- ✓ TensorFlow LSTM backend

Tech Stack & Libraries

Deep Learning

- ▶ TensorFlow / Keras
- ▶ LSTM, CNN, SNN Models
- ▶ GloVe 6B 100d Embeddings

NLP & Data

- ▶ NLTK (stopwords)
- ▶ Keras Tokenizer
- ▶ Pandas / NumPy

Visualization

- ▶ Matplotlib
- ▶ Seaborn

Web App

- ▶ Streamlit
- ▶ Python 3.13

Dataset: IMDB Movie Reviews (50,000 reviews from Kaggle) | Platform: VS Code | Language: Python 3.13

SENSE AI

Conclusion & Future Work

✓ What We Achieved

- ✓ Trained 3 deep learning models on 50K reviews
- ✓ LSTM achieved ~90% training accuracy
- ✓ ~83.5% accuracy on 10,000 test reviews
- ✓ Built & deployed SENSE AI web application
- ✓ End-to-end NLP pipeline implemented

🚀 Future Improvements

- ▶ Integrate BERT / Transformer models
- ▶ Multi-class sentiment (1–5 star ratings)
- ▶ Real-time web scraping from IMDb
- ▶ Mobile-friendly app interface
- ▶ Expand to TV shows & streaming data

Thank you — Questions Welcome!